8.2 Compositions and Inverse Transformations

In Section 1.9 we discussed compositions and inverses of matrix transformations. In this section we extend those ideas to general linear transformations.

One-to-One and Onto

Definition 1

If $T : V \to W$ is a linear transformation, then T is **one-to-one** if T maps distinct vectors in V into distinct vectors in W.

Definition 2

If $T : V \to W$ is a linear transformation, then T is **onto** (or **onto** W) if every vector in W is the image of at least one vector in V.

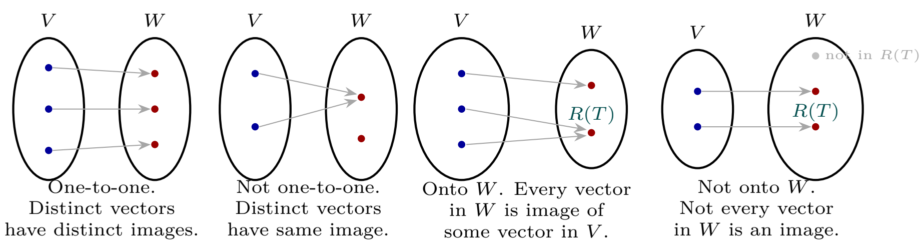

FIGURE 8.2.1

The one-to-one property can also be expressed as:

1. $T : V \to W$ is one-to-one if and only if for each $\mathbf{w}$ in the range of T there is exactly one $\mathbf{v}$ in V with $T(\mathbf{v}) = \mathbf{w}$.

2. $T : V \to W$ is one-to-one if and only if $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$.

Theorem 8.2.1

If $T : V \to W$ is a linear transformation, the following are equivalent:

(a) T is one-to-one.

(b) $\ker(T) = \{\mathbf{0}\}$.

Proof $(a) \Rightarrow (b)$**.** Since T is linear, $T(\mathbf{0}) = \mathbf{0}$. Since T is one-to-one, no other vector maps to $\mathbf{0}$, so $\ker(T) = \{\mathbf{0}\}$.

Proof $(b) \Rightarrow (a)$. Suppose $\ker(T) = \{\mathbf{0}\}$ and $\mathbf{u}, \mathbf{v}$ are distinct vectors in V, so $\mathbf{u} - \mathbf{v} \neq \mathbf{0}$. If $T(\mathbf{u} - \mathbf{v}) = \mathbf{0}$, then $\mathbf{u} - \mathbf{v}$ would be in $\ker(T)$, a contradiction. So $T(\mathbf{u}) - T(\mathbf{v}) = T(\mathbf{u} - \mathbf{v}) \neq \mathbf{0}$, meaning T is one-to-one. ■

EXAMPLE 1 | Rotation Operators on R^2 Are One-to-One and Onto

The linear operator $T : R^2 \to R^2$ rotating each vector through angle θ is one-to-one: distinct vectors rotate into distinct vectors (Figure 8.2.2). It is also onto because every vector in R^2 is a rotation of another vector through θ.

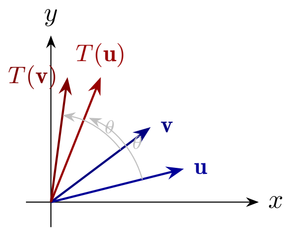

FIGURE 8.2.2 Distinct vectors $\mathbf{u}$ and $\mathbf{v}$ rotate into distinct vectors $T(\mathbf{u})$ and $T(\mathbf{v})$.

EXAMPLE 2 | Orthogonal Projections in R^2 Are Not One-to-One

The operator $T : R^2 \to R^2$ projecting onto the x-axis maps distinct points on any vertical line to the same point on the x-axis, so it is not one-to-one (Figure 8.2.3). It is also not onto because points off the x-axis are not images.

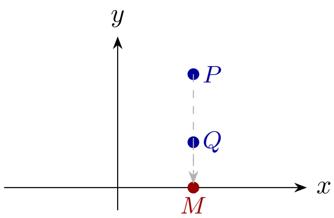

FIGURE 8.2.3 Distinct points P and Q both project to M.

EXAMPLE 3 | Two Transformations That Are One-to-One and Onto

$T_1 : P_3 \to R^4$ and $T_2 : M_{22} \to R^4$ defined by

$$T_1(a + bx + cx^2 + dx^3) = (a, b, c, d) \qquad T_2\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a, b, c, d)$$

are both onto R^4 (any (a, b, c, d) is achievable) and one-to-one (kernels contain only $\mathbf{0}$—verify).

EXAMPLE 4 | A One-to-One Transformation That Is Not Onto

The transformation $T : P_n \to P_{n+1}$, $T(\mathbf{p}) = xp(x)$ from Example 5 of Section 8.1 is one-to-one: if $p(x) \neq q(x)$ they differ in some coefficient, so $T(\mathbf{p}) = xp(x)$ and $T(\mathbf{q}) = xq(x)$ also differ. However, T is not onto because no polynomial in P_n maps to the constant polynomial $1 \in P_{n+1}$ (every image has a zero constant term).

EXAMPLE 5 | Shifting Operators

Let $V = R^\infty$ be the sequence space and define the shifting operators

$$T_1(u_1, u_2, \ldots, u_n, \ldots) = (0, u_1, u_2, \ldots, u_n, \ldots)$$

$$T_2(u_1, u_2, \ldots, u_n, \ldots) = (u_2, u_3, \ldots, u_n, \ldots)$$

(a) T_1 is one-to-one: distinct sequences have distinct images (distinct entries are just shifted). It is not onto: no vector maps to $(1, 0, 0, \ldots)$.

(b) T_2 is onto: any sequence $(u_2, u_3, \ldots)$ is the image of $(0, u_2, u_3, \ldots)$. It is not one-to-one: both $(1, 0, 0, \ldots)$ and $(2, 0, 0, \ldots)$ map to $(0, 0, \ldots)$.

EXAMPLE 6 | Differentiation Is Not One-to-One

(*Calculus Required*)

The differentiation operator $D : C^1(-\infty, \infty) \to F(-\infty, \infty)$, $D(\mathbf{f}) = f'(x)$, is not one-to-one because functions differing by a constant have the same derivative: $D(x^2) = D(x^2 + 1) = 2x$.

When V and W are finite-dimensional with equal dimension, a third condition can be added to Theorem 8.2.1.

Theorem 8.2.2

If V and W are finite-dimensional vector spaces with $\dim(V) = \dim(W)$, and $T : V \to W$ is a linear transformation, the following are equivalent:

(a) T is one-to-one.

(b) $\ker(T) = \{\mathbf{0}\}$.

(c) T is onto $[R(T) = W]$.

Proof. Parts (a) and (b) are equivalent by Theorem 8.2.1. The equivalence of (b) and (c) follows by applying Theorem 8.1.4 with $\dim(V) = n$: since $\text{rank}(T) + \text{nullity}(T) = n$, we have $\text{nullity}(T) = 0$ if and only if $\text{rank}(T) = n = \dim(W)$, which means $R(T) = W$. ■

Remark on dimension. If $\dim(W) < \dim(V)$, then T cannot be one-to-one. If $\dim(V) < \dim(W)$, then T cannot be onto.

Matrix Transformations Revisited

EXAMPLE 7 | Matrix Transformations from R^n to R^m

If $T_A : R^n \to R^m$ is multiplication by an $m \times n$ matrix A:

- If $m < n$: T_A is not one-to-one (nullity > 0) and may or may not be onto.
- If $n < m$: T_A is not onto and may or may not be one-to-one.
- If $m = n$: whether T_A is one-to-one/onto depends on the rank of A.

If A is invertible, T_A is both one-to-one and onto.

Theorem 8.2.3

If $T_A : R^n \to R^m$ is a matrix transformation, then:

(a) T_A is one-to-one if and only if the columns of A are linearly independent.

(b) T_A is onto if and only if the columns of A span R^m.

Proof of (a). By Theorem 8.2.1, T_A is one-to-one iff $\ker(T_A) = \{\mathbf{0}\}$, i.e., A has nullity 0. By Theorem 4.9.2, this means A has rank m columns being linearly independent (full column rank).

Proof of (b). T_A is onto iff $R(T_A) = R^m$, i.e., every $\mathbf{b} \in R^m$ lies in the column space of A (Theorem 4.8.1). ■

Theorem 8.2.4 — Equivalent Statements

If A is an $n \times n$ matrix, the following are equivalent (adding to the prior equivalence list):

(a) A is invertible.

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(c) The reduced row echelon form of A is I_n.

(d) A is a product of elementary matrices.

(e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.

(f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.

(g) $\det(A) \neq 0$.

(h) The column vectors of A are linearly independent.

...

(t) The kernel of T_A is $\{\mathbf{0}\}$.

(u) The range of T_A is R^n.

(v) T_A is one-to-one.

The same subspace of R^n can be viewed three ways:

- **Matrix view:** null space of A
- **System view:** solution space of $A\mathbf{x} = \mathbf{0}$
- **Transformation view:** kernel of T_A

Similarly the same subspace of R^m can be viewed as the column space, the set of consistent right-hand sides, or the range of T_A.

Inverse Linear Transformations

If $T : V \to W$ is one-to-one and $\mathbf{w}$ is any vector in $R(T)$, there is exactly one $\mathbf{v}$ in V with $T(\mathbf{v}) = \mathbf{w}$. This defines the **inverse** $T^{-1} : R(T) \to V$ by $T^{-1}(\mathbf{w}) = \mathbf{v}$.

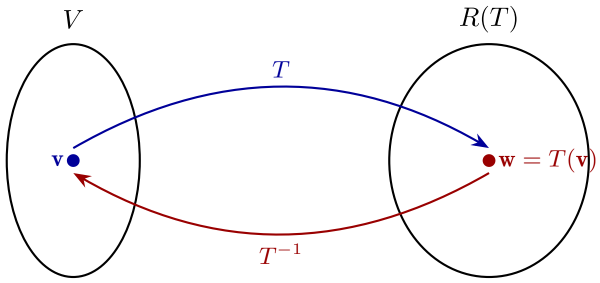

FIGURE 8.2.4 The inverse T^{-1} maps $T(\mathbf{v})$ back to $\mathbf{v}$.

From the definition of T^{-1}, applying T^{-1} then T (or T then T^{-1}) returns the original vector:

$$T^{-1}(T(\mathbf{v})) = T^{-1}(\mathbf{w}) = \mathbf{v} \tag{1}$$

$$T(T^{-1}(\mathbf{w})) = T(\mathbf{v}) = \mathbf{w} \tag{2}$$

so T and T^{-1}, applied in succession in either order, cancel each other. (One can show that $T^{-1} : R(T) \to V$ is itself a linear transformation.)

EXAMPLE 8 | A One-to-One Matrix Transformation

Let $T : R^3 \to R^3$ be defined by $T(x_1, x_2, x_3) = (3x_1 + x_2,\ -2x_1 - 4x_2 + 3x_3,\ 5x_1 + 4x_2 - 2x_3)$. The standard matrix is

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

This matrix is invertible with

$$A^{-1} = \begin{bmatrix} 4 & -2 & -3 \\ -11 & 6 & 9 \\ -12 & 7 & 10 \end{bmatrix}$$

so T is one-to-one and

$$T^{-1}(x_1, x_2, x_3) = (4x_1 - 2x_2 - 3x_3,\ -11x_1 + 6x_2 + 9x_3,\ -12x_1 + 7x_2 + 10x_3)$$

EXAMPLE 9 | An Inverse Transformation

The transformation $T : P_n \to P_{n+1}$, $T(\mathbf{p}) = xp(x)$ (Example 4), is one-to-one but not onto. Its range consists of all polynomials of degree $\leq n + 1$ with zero constant term. The inverse is defined on this range by

$$T^{-1}(c_0x + c_1x^2 + \cdots + c_nx^{n+1}) = c_0 + c_1x + \cdots + c_nx^n$$

For example (with $n \geq 3$): $T^{-1}(2x - x^2 + 5x^3 + 3x^4) = 2 - x + 5x^2 + 3x^3$.

Composition of Linear Transformations

Definition 3

If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, the **composition** of T_2 with T_1, denoted $T_2 \circ T_1$, is the function $U \to W$ defined by

$$(T_2 \circ T_1)(\mathbf{u}) = T_2(T_1(\mathbf{u})) \tag{4}$$

where $\mathbf{u}$ is a vector in U. The domain of T_2 must contain the range of T_1.

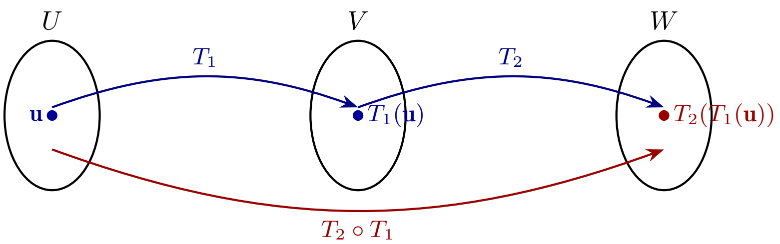

FIGURE 8.2.5 The composition $T_2 \circ T_1$.

Theorem 8.2.5

If $T_1 : U \to V$ and $T_2 : V \to W$ are linear transformations, then $T_2 \circ T_1 : U \to W$ is also a linear transformation.

Proof. For vectors $\mathbf{u}, \mathbf{v}$ in U and scalar c:

$$\begin{aligned}(T_2 \circ T_1)(\mathbf{u}+\mathbf{v}) &= T_2(T_1(\mathbf{u}+\mathbf{v})) = T_2(T_1(\mathbf{u}) + T_1(\mathbf{v})) \\ &= T_2(T_1(\mathbf{u})) + T_2(T_1(\mathbf{v})) = (T_2 \circ T_1)(\mathbf{u}) + (T_2 \circ T_1)(\mathbf{v}) \\ (T_2 \circ T_1)(c\mathbf{u}) &= T_2(T_1(c\mathbf{u})) = T_2(cT_1(\mathbf{u})) = cT_2(T_1(\mathbf{u})) = c(T_2 \circ T_1)(\mathbf{u}) \quad \blacksquare\end{aligned}$$

EXAMPLE 10 | Composition of Linear Transformations

Let $T_1 : P_1 \to P_2$ and $T_2 : P_2 \to P_2$ be given by $T_1(p(x)) = xp(x)$ and $T_2(p(x)) = p(2x+4)$. Then $(T_2 \circ T_1)(p(x)) = T_2(xp(x)) = (2x+4)p(2x+4)$. For $p(x) = c_0 + c_1 x$:

$$(T_2 \circ T_1)(c_0 + c_1 x) = (2x+4)(c_0 + c_1(2x+4)) = c_0(2x+4) + c_1(2x+4)^2$$

EXAMPLE 11 | Composition with the Identity Operator

If $T : V \to V$ is any linear operator and $I : V \to V$ is the identity operator, then for all $\mathbf{v}$ in V:

$$(T \circ I)(\mathbf{v}) = T(I(\mathbf{v})) = T(\mathbf{v}) \qquad (I \circ T)(\mathbf{v}) = I(T(\mathbf{v})) = T(\mathbf{v})$$

Thus $T \circ I$ and $I \circ T$ are both the same as T:

$$T \circ I = T \qquad \text{and} \qquad I \circ T = T \tag{5}$$

Compositions extend to three or more transformations. If $T_1 : U \to V$, $T_2 : V \to W$, $T_3 : W \to Y$, then

$$(T_3 \circ T_2 \circ T_1)(\mathbf{u}) = T_3(T_2(T_1(\mathbf{u}))) \tag{6}$$

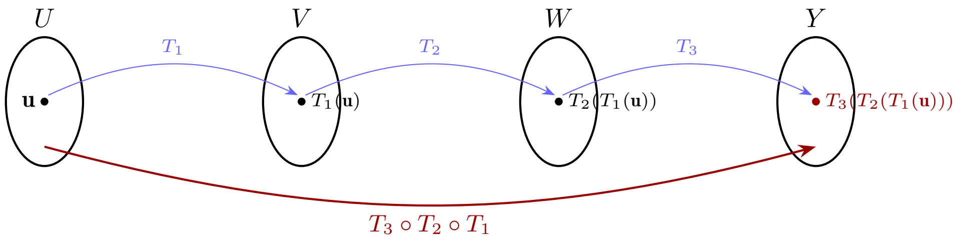

FIGURE 8.2.6 Composition of three linear transformations.

Composition of One-to-One Linear Transformations

Theorem 8.2.6

If $T_1 : U \to V$ and $T_2 : V \to W$ are one-to-one linear transformations, then:

(a) $T_2 \circ T_1$ is one-to-one.

(b) $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$.

Proof of (a). If $\mathbf{u}$ and $\mathbf{v}$ are distinct vectors in U, then $T_1(\mathbf{u}) \neq T_1(\mathbf{v})$ (since T_1 is one-to-one), and then $T_2(T_1(\mathbf{u})) \neq T_2(T_1(\mathbf{v}))$ (since T_2 is one-to-one). So $(T_2 \circ T_1)(\mathbf{u}) \neq (T_2 \circ T_1)(\mathbf{v})$. ■

Proof of (b). Let $\mathbf{u} = (T_2 \circ T_1)^{-1}(\mathbf{w})$ for any $\mathbf{w}$ in the range of $T_2 \circ T_1$. Then $(T_2 \circ T_1)(\mathbf{u}) = \mathbf{w}$, i.e., $T_2(T_1(\mathbf{u})) = \mathbf{w}$. Taking T_2^{-1} gives $T_1(\mathbf{u}) = T_2^{-1}(\mathbf{w})$, and then taking T_1^{-1} gives $\mathbf{u} = T_1^{-1}(T_2^{-1}(\mathbf{w})) = (T_1^{-1} \circ T_2^{-1})(\mathbf{w})$. ■

Part (b) generalizes: the inverse of a composition is the composition of the inverses in *reverse* order. For three transformations:

$$(T_3 \circ T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1} \circ T_3^{-1} \tag{8}$$

For matrix transformations this gives $(T_{BA})^{-1} = T_{A^{-1}B^{-1}}$ and $(T_{CBA})^{-1} = T_{A^{-1}B^{-1}C^{-1}}$, or equivalently:

$$(BA)^{-1} = A^{-1}B^{-1} \qquad \text{and} \qquad (CBA)^{-1} = A^{-1}B^{-1}C^{-1} \tag{9}$$

Exercise Set 8.2

In Exercises 1–2, determine whether the stated matrix operator is one-to-one.

1. **a.** The orthogonal projection onto the x-axis in R^2.
   b. The reflection about the y-axis in R^2.
   c. The reflection about the line $y = x$ in R^2.

2. **a.** A rotation about the z-axis in R^3.
   b. A reflection about the xy-plane in R^3.
   c. An orthogonal projection onto the xz-plane in R^3.

In Exercises 3–4, determine whether T is one-to-one by finding its kernel and applying Theorem 8.2.1.

3. **a.** $T : R^2 \to R^2$, $T(x, y) = (y, x)$.
   b. $T : R^2 \to R^3$, $T(x, y) = (x, y, x + y)$.
   c. $T : R^3 \to R^2$, $T(x, y, z) = (x + y + z,\ x - y - z)$.

4. **a.** $T : R^2 \to R^3$, $T(x, y) = (x - y,\ y - x,\ 2x - 2y)$.
   b. $T : R^2 \to R^2$, $T(x, y) = (0, 2x + 3y)$.
   c. $T : R^2 \to R^2$, $T(x, y) = (x + y,\ x - y)$.

In Exercises 5–6, use Theorem 8.2.3 to determine whether T_A is one-to-one, onto, both, or neither.

5. **a.** $A = \begin{bmatrix} 1 & -2 \\ 2 & -4 \\ -3 & 6 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 1 & 3 & 1 & 7 \\ 2 & 7 & 2 & 4 \\ -1 & -3 & 0 & 0 \end{bmatrix}$

7. Use the given information to determine whether T is one-to-one.
   a. $T : V \to W$; $\text{nullity}(T) = 0$.
   b. $T : V \to W$; $\text{rank}(T) = \dim(V)$.
   c. $T : V \to W$; $\dim(W) < \dim(V)$.

8. Use the given information to determine whether the linear operator $T : V \to V$ is one-to-one, onto, both, or neither.
   a. $T : V \to V$; $\text{nullity}(T) = 0$.
   b. $T : V \to V$; $\text{rank}(T) < \dim(V)$.

11. Let $\mathbf{a}$ be a fixed vector in R^3. Does $T(\mathbf{v}) = \mathbf{a} \times \mathbf{v}$ define a one-to-one linear operator on R^3? Explain.

12. Let E be a fixed 2×2 elementary matrix. Does $T(A) = EA$ define a one-to-one linear operator on M_{22}? Explain.

In Exercises 13–14, use Theorem 8.2.3 to determine whether multiplication by A is one-to-one, onto, both, or neither.

13. **a.** $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 1 & 3 & 1 & 1 \\ 1 & 4 & 0 & 0 \end{bmatrix}$ **c.** $A = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$

15. In words, describe the inverse of each operator.
    a. Reflection about the x-axis in R^2.
    b. Rotation about the origin through $\pi/4$ in R^2.

17. Use matrix inversion to confirm the stated result in R^2.
    a. The inverse of a reflection about $y = x$ is a reflection about $y = x$.
    b. The inverse of a rotation through θ is a rotation through $-\theta$.

19. Let $T : P_1 \to R^2$ be defined by $T(p(x)) = (p(0), p(1))$.
    a. Find $T(1 - 2x)$.
    b. Show T is a linear transformation.
    c. Show T is one-to-one.
    d. Find $T^{-1}(2, 3)$ and sketch its graph.

20. Determine whether $T : R^n \to R^n$ is one-to-one; if so, find T^{-1}.
a. $T(x_1, \ldots, x_n) = (0, x_1, x_2, \ldots, x_{n-1})$.
b. $T(x_1, x_2, \ldots, x_n) = (x_n, x_{n-1}, \ldots, x_2, x_1)$.

In Exercises 23–24, compute $(T_2 \circ T_1)(x, y)$.

23. $T_1(x, y) = (2x, 3y)$, $T_2(x, y) = (x - y,\ x + y)$.

24. $T_1(x, y) = (2x, -3y, x + y)$, $T_2(x, y, z) = (x - y,\ y + z)$.

25. Let $T_1 : P_2 \to P_2$ and $T_2 : P_2 \to P_3$ be given by $T_1(p(x)) = p(x + 1)$ and $T_2(p(x)) = xp(x)$. Find $(T_2 \circ T_1)(a_0 + a_1 x + a_2 x^2)$.

26. Let $T_1 : P_n \to P_n$ and $T_2 : P_n \to P_n$ be defined by $T_1(p(x)) = p(x - 1)$ and $T_2(p(x)) = p(x + 1)$. Find $(T_1 \circ T_2)(p(x))$ and $(T_2 \circ T_1)(p(x))$.

27. Let $T_1 : M_{22} \to R$ and $T_2 : M_{22} \to M_{22}$ be defined by $T_1(A) = \mathrm{tr}(A)$ and $T_2(A) = A^T$.
a. Find $(T_1 \circ T_2)(A)$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
b. Can you find $(T_2 \circ T_1)(A)$? Explain.

32. Let $T_1 : R^2 \to R^2$ and $T_2 : R^2 \to R^2$ be the operators $T_1(x, y) = (x + y,\ x - y)$ and $T_2(x, y) = (2x + y,\ x - 2y)$.
a. Show T_1 and T_2 are one-to-one.
b. Find $T_1^{-1}(x, y)$, $T_2^{-1}(x, y)$, and $(T_2 \circ T_1)^{-1}(x, y)$.
c. Verify $(T_2 \circ T_1)^{-1} = T_1^{-1} \circ T_2^{-1}$.

35. Let $T : R^3 \to R^3$ be orthogonal projection onto the xy-plane. Show $T \circ T = T$.

38. (*Calculus required*) Let $D : P_n \to P_{n-1}$ by $D(p(x)) = p'(x)$ and $J : P_{n-1} \to P_n$ by $J(p(x)) = \int_0^x p(t)\, dt$.
a. Show D and J are linear.
b. Explain why J is not the inverse of D.
c. Can the domains/codomains of D and J be restricted so they are inverse linear transformations?

Working with Proofs

44. Prove: if there is an onto linear transformation $T : V \to W$, then $\dim(V) \geq \dim(W)$.

45. Prove: if $T : V \to W$ is a one-to-one linear transformation, then $T^{-1} : R(T) \to V$ is also a one-to-one linear transformation.

46. Using Formula (6), prove:
a. $T_3 \circ T_2 \circ T_1$ is a linear transformation.
b. $T_3 \circ T_2 \circ T_1 = (T_3 \circ T_2) \circ T_1$.
c. $T_3 \circ T_2 \circ T_1 = T_3 \circ (T_2 \circ T_1)$.

47. Let V and W be finite-dimensional and $T : V \to W$ linear. Prove:
a. If $\dim(V) > \dim(W)$, then T cannot be one-to-one.
b. If $\dim(V) < \dim(W)$, then T cannot be onto.